DEEPTHI CHOWDARY EDARA

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PROFESSIONAL SUMMARY:

- **AI/ML** Engineer & **Data Scientist** with **5+** years of experience in designing, deploying, and optimizing machine learning models and AI systems.
- Expertise in **MLOps**, **AI observability**, and **automation frameworks**, ensuring seamless AI deployment and performance monitoring.
- Skilled in **LLMs, Generative AI**, **RAG**, **Transformers, Deep Learning, NLP, and Computer Vision**, leveraging cutting- edge AI advancements.
- Proficient in Python, R, SQL, TensorFlow, Py Torch, and various **ML libraries** such as Scikit-learn, XG Boost, and Light GBM.
- Strong experience with cloud platforms **AWS** (Sage Maker, Redshift, Lambda, Bedrock) and **Azure** (Synapse, Data Factory) for scalable AI solutions.
- Implemented **CI/CD** pipelines (GitHub Actions, Jenkins, AWS Code Pipeline) for AI model integration and deployment.
- Extensive experience in **Vector Databases** (Pinecone, FAISS) for optimizing high-dimensional data retrieval and AI-powered search systems.
- Developed AI-powered automation for **testing, drift detection, and retraining**, ensuring continuous model performance improvements.
- Built real-time AI applications such as **Conversational AI platforms**, NLP-based automation, and intelligent service recommendation systems.
- Designed and optimized scalable **database architectures** (SQL, PostgreSQL, MongoDB, Snowflake) for data-driven decision-making.
- Developed AI-driven business insights and BI dashboards (**Power BI, Tableau, Quick Sight**) for data visualization and decision-making.
- Proven ability to lead AI and Data Science projects across industries such as **Finance, Healthcare, and Telecommunications.**

TECHNICAL SKILLS:

Category	**Technologies**
Languages	Python, JAVA, SQL, R
AI Domains	Generative AI, Multi-Modality RAG, Agentic RAG, Deep-seek, Transformers, LLMs (GPT-3, GPT- 3.5, GPT-4.0, Bedrock Models), Computer Vision(SAM,Object Detection) NLP, Reinforcement Learning, Prompt Engineering.
Databases	SQL Server, MySQL, PostgreSQL, MongoDB, Snowflake, Vector Databases (Pinecone, FAISS)
Deep Learning	CNN, RNN, ANN, Vision Transformers, NLTK, Spacy, BERT

Cloud & DevOps Platforms	AWS (S3, Redshift, EMR, Lambda, EC2, CloudWatch, SageMaker, Bedrock, SageMaker Pipelines, Step Functions, Glue, Feature Store, Model Registry, Code Pipeline, Fargate), Azure (Synapse, Data Factory, Machine Learning, HDInsight, Event Hubs, Stream Analytics, Key Vault), Docker, Kubernetes (EKS), Terraform, CI/CD (GitHub Actions, Jenkins). MLOps & Model Deployment: Kubeflow, MLflow, Prometheus, Evidently AI, Arize AI, AWS Model Registry, Model Drift Detection, Automated Model Retraining.
ML Libraries/Frameworks	TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, SVM, Random Forest
Analytical Techniques	Regression , Clustering (K-means), PCA, LDA, Time Series Forecasting (ARIMA)
ETL & Data Engineering Tools	AWS Glue, Apache Airflow, dbt.
Monitoring & MLOps	AWS CloudWatch
Version Control & Collaboration	Git, GitHub, GitLab, Bitbucket
Business Intelligence & Reporting	Power BI, Tableau

Education:

University of Memphis, Memphis, TN, USA

Master's in information systems (Top 1% of class)

Relevant Courses: - Data Mining & Predictive Analytics, Machine Learning for Business Intelligence, Database Management Systems, Cloud Computing & Distributed Systems, Information Security & Risk Management.

Vignan University, Vadlamudi, Guntur, A.P, India

B.Tech. in Computer Science Engineering (Top 1% of class)

Relevant Courses: - Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning, Cloud Computing.

Certifications:

PROFESSIONAL EXPERIENCE:

AI Engineer – Fedex, Memphis, TN **May 2025 – Present**

Project: AI-Based Smart Package Risk & Damage Prediction System

Roles & Responsibilities:

- Designed and implemented an **AI-powered logistics intelligence platform** to predict delivery delays and optimize shipment routing across large-scale transportation networks**.**
- Built **machine learning models** using historical shipment data, weather conditions, traffic patterns, and hub congestion metrics to estimate delivery ETA and delay risks.
- Implemented **real-time data ingestion and inference pipelines** to process live shipment scans and GPS events for proactive exception handling.
- Developed a **route optimization and risk scoring engine** to dynamically recommend alternate paths during operational disruptions.

- Integrated the AI system with **logistics and tracking APIs** to support automated decision-making for dispatch and operations teams.
- Deployed **end-to-end MLOps pipelines** with model versioning, monitoring, and automated retraining to ensure scalability and reliability.

Environment Python, XGBoost, Kafka, Spark Streaming, AWS , Azure, Docker, Kubernetes.
Methodologies: Machine Learning, Real-Time Analytics, Route Optimization, MLOps, Scalable AI Systems.

Machine Learning Engineer – Bank of America, Memphis, TN **May 2024 -April 2025**
Project: AI-Powered Financial Assistant System
Roles & Responsibilities:

- Designed and implemented a **conversational AI platform** to assist banking customers with account management, credit score insights, and proactive fraud alerts.
- Integrated **LLMs (OpenAI GPT, AWS Bedrock)** to provide personalized financial guidance and explain credit components in simple language.
- Developed **secure chatbot interactions** with banking APIs enabling self-service actions such as password resets, statement retrieval, and card blocking.
- Built a **semantic search layer** using **FAISS and PGVector** to enable contextual retrieval of transaction histories, compliance rules, and dispute policies, improving response time by **90%.**
- Enhanced **security and compliance** through multi-factor authentication, JWT-based session continuity, and role-based access controls.
- Conducted **A/B testing on conversation flows**, increasing user engagement and completion of safety actions by 35%.

Environment: OpenAI GPT, AWS Bedrock, FAISS, PGVector, Python, Banking APIs.
Methodologies: Agile, Conversational AI, Semantic Search ,Secure API Integration, Customer Experience Optimization.

Data Scientist - Cognizant, Hyderabad, India **April 2022 – Nov 2023**
Project: AI-Powered Demand Forecasting & Inventory
Roles & Responsibilities:

- Built a **predictive demand forecasting model** for a retail client using historical sales data, promotions, seasonality, and external economic indicators.
- Implemented **time-series models (ARIMA, Prophet, LSTM)** to improve forecast accuracy and handle complex seasonal trends.
- Engineered features from multiple sources, including point-of-sale, inventory logs, and regional trends, to enhance model performance.
- Developed an **inventory optimization module** to recommend stock levels at warehouse and store levels, reducing overstock and stock-outs.
- Built an **end-to-end data pipeline** for data ingestion, preprocessing, model training, validation, and deployment using Python and cloud services (AWS/Azure).
- Created **interactive dashboards** for business users to visualize forecasts, inventory recommendations, and performance metrics in real time.
- Monitored model performance and retrained models periodically to account for **seasonal changes and demand fluctuations**.
- Achieved **10–15% improvement in forecast accuracy** and **20% reduction in excess inventory**, optimizing supply chain operations.
- Collaborated with **cross-functional teams**, including operations, supply chain, and IT, to ensure seamless integration into business workflows.
- Ensured **scalability, reproducibility, and maintainability** by building modular code, standardized workflows, and automated reporting.

Environment: Python, Pandas, NumPy, Scikit-learn, ARIMA, LSTM, Prophet, AWS/Azure, Tableau/Power BI.

Methodologies: Time-Series Forecasting, Feature Engineering, Data Pipeline Automation, Inventory Optimization, Business Impact Analysis.

Data Analyst – Wipro, Hyderabad, India **Jan 2021 – Feb 2022**

Project: Healthcare Data Quality & Analytics

Roles & Responsibilities:

- Conducted **data quality** assessments on regulated healthcare datasets to detect inconsistencies impacting compliance, billing accuracy, and reporting reliability.
- Developed **Tableau dashboards** tracking KPIs such as patient classification metrics, billing exception rates, reimbursement timelines, and financial anomaly patterns for executive leadership.
- Led **Python-based statistical modeling** to monitor healthcare operational performance and identify root causes of inefficiencies, enabling data-driven corrective actions.
- Processed and analyzed **large-scale pharmaceutical datasets** using **Python and SQL**, uncovering patterns that increased **drug development** process efficiency **by 15%**.
- Collaborated closely with **cross-functional stakeholders** including clinical, finance, compliance, and product teams to align analytics insights with strategic business objectives—supporting **two successful pharmaceutical product launches**.
- Designed interactive **dashboards in Power BI and Tableau** to display real-time KPIs and forecasting metrics, improving speed and accuracy of decision-making for stakeholders.
- **Streamlined ETL and validation workflows** to enhance data quality, minimizing audit discrepancies and improving reporting confidence across teams.
- **Utilized SQL and DAX for data modeling and forecasting**, improving visibility into market demand trends, promotional effectiveness, and revenue projections

Environment Python, SQL, Tableau, Power BI, DAX, SharePoint, Excel, Data Governance Frameworks, Healthcare Data Standards, ETL Pipelines, Agile.

Methodologies: Data Quality & Data Governance, ETL / Data Pipeline Optimization, KPI Framework Design, Statistical Modeling, A/B Testing, Agile, Dashboard Development.

Awards & Achievements

- **Employee Excellence Award** for designing and deploying an AI-driven fraud detection platform, reducing call center load by 45%
- **Innovation Recognition** for building a real-time conversational AI assistant for credit risk advisory and proactive fraud alerts
- Awarded for **driving measurable business impact** through AI/ML pipelines, improving fraud detection accuracy by 28%